ARRESTSHIELD - HONEY

ML Status and PC Testing Guide

What is complete, what remains for the full project, and exactly how to test the trained ML system on this Windows PC.

ML COMPONENT	FULL PRODUCT
Complete	**Not complete**
Trained, tested, documented, and available through a local text/audio API.	Honeypot, RAG, UI, threat-intelligence storage, and full integration remain.

DATASET	AUTOMATED TESTS	ARTIFACT AUDIT
52,206	**68 / 68**	**28 / 28**
validated conversations	passed locally	hash and policy checks

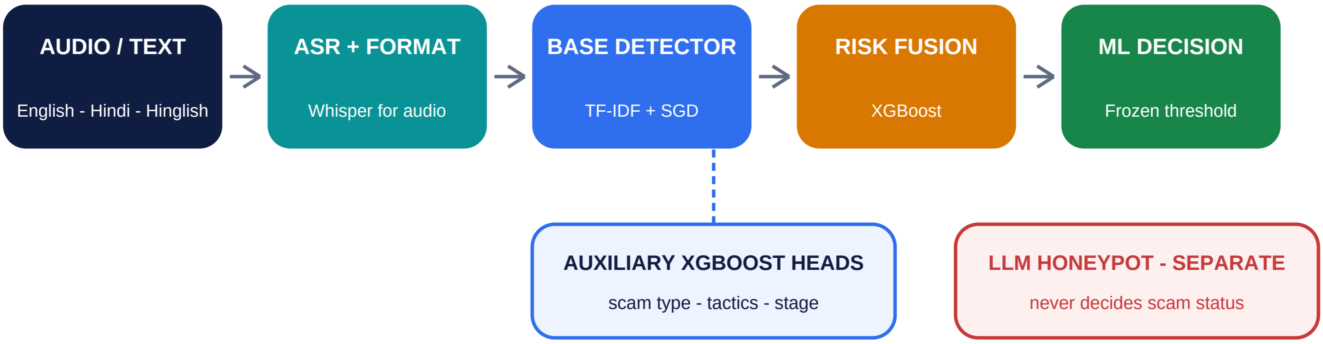
Important status
The ML engineering build is complete as a **research prototype**. It is not production-ready, and it is not the complete ArrestShield-Honey application.

Project folder: **C:\ArrestShield**
Current Git milestone: **2a1e0dd - Complete trained ArrestShield ML pipeline**

STATUS AT A GLANCE

What is complete - and what is not

One-sentence answer
The complete **ML subsystem** is working locally; the broader application around it still needs to be built.



The deployed decision path

- Audio is transcribed locally with multilingual Whisper-tiny; text input skips ASR.
- A class-weighted SGD detector produces the base scam probability.
- XGBoost risk fusion combines the base score with lexical and privacy-aware entity features.
- Separate XGBoost heads report scam type, stage, and supported tactics; they cannot change the scam decision.
- A frozen threshold produces the research decision. The LLM honeypot is outside this path and remains disabled.

Why the word 'research' matters
The API may return **is_scam: true**, but **production_eligible remains false**. Current data cannot justify a real-world accuracy or deployment claim.

COMPLETED WORK

Completed ML components

Everything below is implemented and has local evidence in the repository.

Area	What is complete	Evidence / outcome
Dataset pipeline	Registry, licensed-source ingestion, normalization, deduplication, canonical schema, grouped splits	52,206 conversations; 615,084 turns; zero split-group leakage
Baseline models	Word/character TF-IDF with class-weighted SGD and SVD-XGBoost	Seeds 17, 42, and 93 trained
Multi-task ML	XGBoost heads for scam type, current stage, and 9 supported tactics	Three-seed compact artifact; 1.92 MiB head bundle
Risk fusion	XGBoost over out-of-fold base scores plus lexical/entity signals	Trained and loaded by the local API
Early detection	Causal prefix windows, stable-turn latency, threshold selection at fixed FPR	Future labels cannot leak into earlier prefixes
ASR	Local Whisper-tiny transcription, file checks, duration checks, temporary deletion	Real FLAC request passed end to end
Entity extraction	UPI, phone, email, URL, account/Aadhaar/OTP candidates, amount, case, authority, app	Sensitive values redacted by default
Inference API	Health, model information, text detection, and audio detection endpoints	Runs locally at 127.0.0.1:8000
Testing and integrity	Logic tests, API contracts, artifact hashes, policy-boundary checks	68 tests passed; 28/28 verifier checks passed
Packaging	Offline model paths, localhost policy, non-root Docker definition	Local research deployment prepared
LLM boundary	False flags in config, manifests, reports, verifier, and responses	LLM never labels, scores, thresholds, or decides

Deployed model family
The laptop-safe deployed system uses **SGD + XGBoost**. The causal multilingual transformer code and exact-resume mechanism are implemented for future GPU comparison, but no completed transformer quality claim is made.

RESULTS AND LIMITATIONS

What the current numbers mean

SCAM TYPE MACRO-F1	STAGE MACRO-F1	SUPPORTED TACTICS F1
0.853	0.839	0.707
validation mean across 3 seeds	validation mean across 3 seeds	validation mean across 9 heads

Rigorous controls already applied

- Training, validation, and test partitions are grouped by conversation rather than individual turns.
- Thresholds are chosen using validation hard negatives at a pre-registered maximum 5% false-positive rate.
- Three seeds are reported to expose random-seed variation.
- Strict leave-one-source-out audits refit the representation and models without the held source.
- High mixed-source numbers are explicitly treated as supporting evidence, not real-world performance.

Why production promotion is blocked

Limitation	Current evidence	Effect
Positive-label quality	All 2,256 positives are source-silver; 71% are synthetic	Cannot claim real-call accuracy
Source shortcuts	Strict source-macro FPR: SGD 11.15%, XGBoost baseline 7.02%, fusion 27.71%	All exceed the 5% gate
Worst-source behavior	Fusion worst-source FPR is 80.95%	Strong domain-shift risk
Human-gold set	0 of 150 independently annotated conversations collected	Final promotion gate unavailable
Hindi/Hinglish audio	Representative consented validation set is uncollected	ASR backend cannot be promoted
Sparse tactics	Six taxonomy labels have zero positive causal training windows	Returned as unavailable, not false

Correct interpretation
The software pipeline is complete and testable. The scientific evidence is not yet strong enough for public deployment, automatic call intervention, or a claimed production accuracy percentage.

REMAINING WORK

What remains for the full ArrestShield-Honey project

Workstream	What still needs to be built	Dependency / completion signal
LLM fake-victim honeypot	Safe persona, adaptive multi-turn response policy, refusal/safety rules, conversation memory	Separate service; never inserted into ML decision
RAG knowledge layer	Curated scam-pattern knowledge base, retrieval pipeline, grounded response generation	Cited sources, retrieval tests, prompt-injection controls
ML-to-honeypot handoff	Signed event contract, eligibility policy, retries, idempotency, audit trail	Enable only after model promotion and policy approval
Threat-intelligence extraction	Structured entities, indicators, evidence spans, confidence, deduplication	Validated schema and review workflow
Storage and dashboard	Case database, searchable incidents, analytics, investigator-facing UI	Access control, retention, privacy, export
Frontend demo	Upload/text form, live result view, model-status banner, redacted evidence	End-to-end demo usable without Swagger
Telecom/live-call integration	Streaming audio, turn segmentation, latency budget, consent/legal review	Out of current prototype scope
Production security	Authentication, TLS, rate limits, upload scanning, logging, monitoring, rollback	Security and privacy review passed
Human evaluation data	150 independently annotated frozen conversations plus representative Hindi/Hinglish audio	Leakage checks and promotion gates pass
End-to-end validation	Load, failure recovery, red-team, safety, latency, and human usability testing	Documented acceptance report

Recommended order

PHASE 1	PHASE 2	PHASE 3
Data	**Honeypot**	**Product**
human-gold and Hinglish audio	LLM/RAG as separate service	handoff, UI, storage, deployment

Do not connect the honeypot yet
The checked-in policy correctly blocks handoff while the detector is research-only. Build and test the honeypot independently, but keep automatic engagement disabled until the data and source-generalization gates pass.

TEST ON THIS PC

Quick local test - no retraining required

Good news
The virtual environment, datasets, model artifacts, Whisper files, Python 3.10.11, and FFmpeg are already present on this PC. Do **not** rerun training just to test the system.

1 **Open PowerShell**
Press Start, search for **PowerShell**, and open it. Administrator mode is not required.

GO TO THE PROJECT

```
Set-Location C:\ArrestShield
```

2 **Verify the exact saved artifacts**
This recalculates dataset/model hashes and checks the no-LLM and disabled-handoff policies.

RUN VERIFIER

```
.\.venv\Scripts\python.exe scripts\verify_local_artifacts.py
```

Expected result:

EXPECTED OUTPUT

```
{
  "checks": 28,
  "passed": 28,
  "failed": 0,
  "skipped": 0
}
```

3 **Run the automated suite**
This checks the data rules, early detection logic, XGBoost/multi-task contracts, ASR validation, entity redaction, API behavior, artifact reading, and LLM boundary.

RUN TESTS

```
.\.venv\Scripts\python.exe -m pytest -q
```

Expected final line: **68 passed**. Pytest cache permission warnings can be ignored if all 68 tests pass.

If a command says Python or a package is missing
Make sure the command uses **.\.venv\Scripts\python.exe**, not the system-wide **python** command. From this PC's existing project folder, no dependency installation should be needed.

INTERACTIVE API TEST

Start the service and test text detection

1

Start the local API
Run this in PowerShell and keep the window open while testing. First startup can take around 15-30 seconds while the local artifacts load.

TERMINAL 1

```
Set-Location C:\ArrestShield
.\.venv\Scripts\python.exe scripts\run_api.py
```

Wait until the terminal shows that Uvicorn is running on **http://127.0.0.1:8000**.

2

Open the interactive test page
In Chrome or Edge, open **http://127.0.0.1:8000/docs**. This Swagger page lets you test without writing code.

3

Check service health
Expand **GET /healthz**, click **Try it out**, then **Execute**. Expect status **200**, status **ok**, and **llm_used_for_detection: false**.

4

Check the loaded model
Run **GET /v1/model**. Confirm **multitask_backend: classical**, **multitask_auxiliary_loaded: true**, and research-only status.

5

Submit a scam-like text
Expand **POST /v1/detect/text**, click **Try it out**, replace the request body with the JSON below, and click **Execute**.

REQUEST BODY

```
{
  "conversation_id": "manual-test-001",
  "turns": [
    {"speaker_role": "caller", "text": "Main CBI officer bol raha hoon."},
    {"speaker_role": "caller", "text": "Kisi ko mat batana. safe@ybl par abhi paise bhejo."}
  ],
  "include_sensitive_entities": false
}
```

INSPECT RESULTS

Read the response and test other inputs

Check the safety and model-status fields as carefully as the predicted label.

Expected fields to inspect

Response field	Expected observation
is_scam / scam_score	Likely true with a high research score for this explicit example
decision_source	trained_xgboost_risk_fusion
auxiliary_signals.scam_type	Likely digital_arrest
entities	UPI appears as ***@ybl because redaction is enabled
production_eligible	false
honeypot.invoked	false
llm_used_for_detection	false

A useful detector test includes both suspicious and ordinary conversations. Do not judge it using only one dramatic example.

Normal-text test

POST /V1/DETECT/TEXT

```
{
  "conversation_id": "manual-test-normal",
  "turns": [
    {"speaker_role": "caller", "text": "Your parcel will arrive tomorrow between 2 PM and 4 PM."}
  ],
  "include_sensitive_entities": false
}
```

The expected result is generally **is_scam: false**, but this remains a research model. Individual examples are demonstrations, not an accuracy evaluation.

Audio test through Swagger

- Open **POST /v1/detect/audio** and click **Try it out**.
- Choose a local **.wav, .flac, .mp3, .m4a, or .ogg** file. Use only audio you are authorized to process.
- Set **language_hint** to **en** or **hi**; leave sensitive-entity output disabled.
- Click **Execute**. CPU transcription can take about one minute for a short clip.
- Confirm an ASR transcript is returned, the temporary path says **temporary_upload_deleted**, and LLM/honeypot flags remain false.

Existing known-good audio smoke
The repository already passed a real local FLAC request. Whisper produced a full English transcript, the ML system returned non-scam, and the temporary upload was deleted.

TROUBLESHOOTING

Finish the test and record what happened

Common issues

Symptom	What to do
Browser cannot open /docs	Keep the API PowerShell window open; wait for the Uvicorn running message; use 127.0.0.1:8000, not a public address
Port 8000 already in use	Close the older API terminal/process, then start the command again
Audio extension rejected	Convert to WAV, FLAC, MP3, M4A, or OGG; verify FFmpeg is available
Audio is slow	Expected on CPU; Whisper loading and decoding are the slowest local path
Pytest cache warning	Ignore only if the final result still says 68 passed
Model returns an unexpected label	Record the example as an error case; do not manually adjust the threshold from one sample
Need to stop the API	Return to Terminal 1 and press Ctrl+C

Record manual test cases

For a meaningful professor/demo report, keep a small test log without changing the frozen model threshold. Suggested columns:

Field	What to record
Test ID and input	A non-sensitive identifier and the exact text, or authorized audio filename
Expected outcome	Why you expected scam or non-scam before looking at the model
Model output	is_scam, score, type, stage, available tactics, and decision source
Safety checks	Redaction status, production_eligible, honeypot.invoked, and LLM flag
Observation	Correct, false positive, false negative, ambiguous, or ASR transcription issue

Testing success checklist
Verifier: 28/28 - tests: 68 passed - /healthz: 200 - model backend: classical - suspicious and normal text requests complete - audio transcript returns - redaction works - LLM and honeypot remain false.

Privacy reminder
Use only conversations and audio you are authorized to process. Keep sensitive-entity output disabled for ordinary testing, and do not commit private call recordings or raw personal data to GitHub.

For detailed technical evidence, see **docs/COMPLETION_AUDIT.md**, **docs/INFERENCE_API.md**, **reports/classical_multitask_v1/RESULTS.md**, and **reports/verification_v1/verification.json** inside C:\ArrestShield.